LAB 5 : Azure Linux VM → Custom Image → New VM

Create VM → Configure → Capture Image → Azure Compute Gallery → Create VM from Image

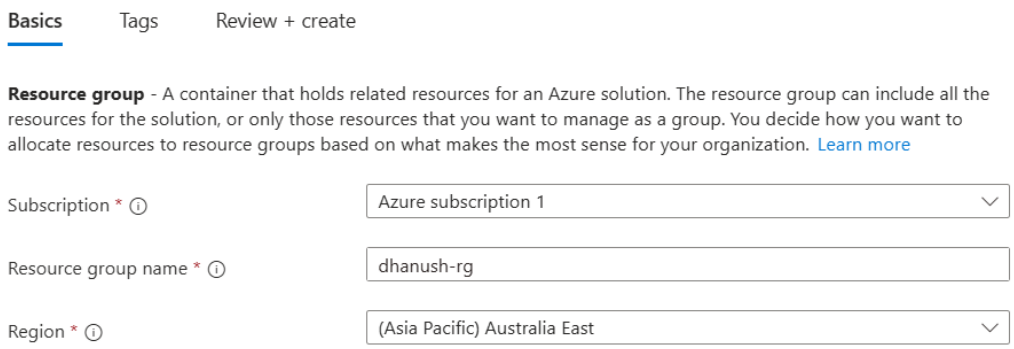
🏷️ Mandatory Naming Convention

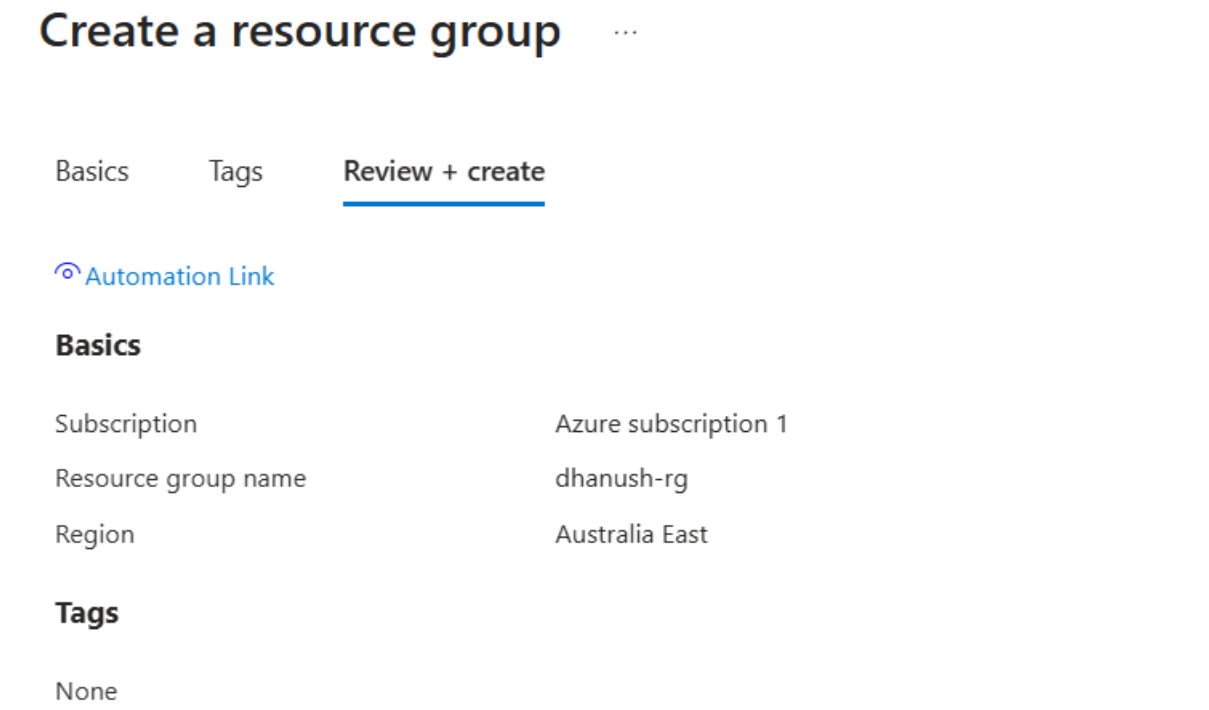
Replace **studentname** with your name (example: balaji).

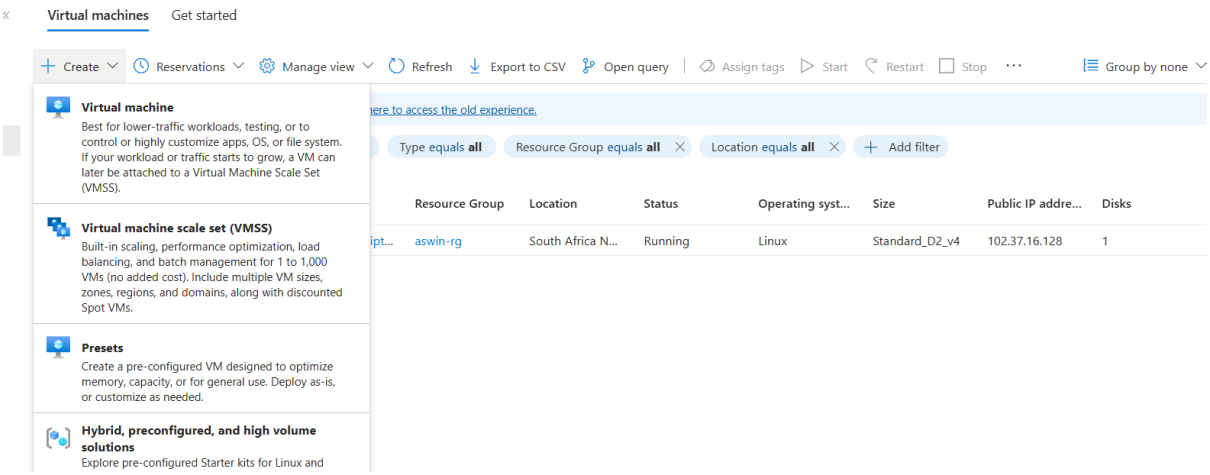
Resource	Name
Resource Group	studentname-rg
Source VM	studentname-vm
Azure Compute Gallery	studentname-acg
Image Definition	studentname-img-def
Image Version	1.0.0
VM from Image	studentname-vm-from-img
Virtual Network	studentname-vnet
Subnet	studentname-subnet

Step 1: Create Resource Group (GUI)

1. Login to **Azure Portal**
2. Search → **Resource groups**
3. Click **Create**
4. Enter:
    - Resource group: studentname-rg
    - Region: **East US**
5. Click **Review + Create → Create**







Step 2: Create Linux Virtual Machine (GUI)

1. Search → **Virtual Machines**
2. Click **Create → Azure virtual machine**

Basics Tab

Fill in the following:

- Resource group: studentname-rg
- Virtual machine name: studentname-vm
- Region: East US
- Availability options: No infrastructure redundancy required
- Security type: Standard
- Image: **RHEL / Rocky / CentOS / Ubuntu**
- Size: Standard_B2s
- Username: azureuser
- Authentication type: SSH public key (or) user & password
- SSH key: Generate new key pair

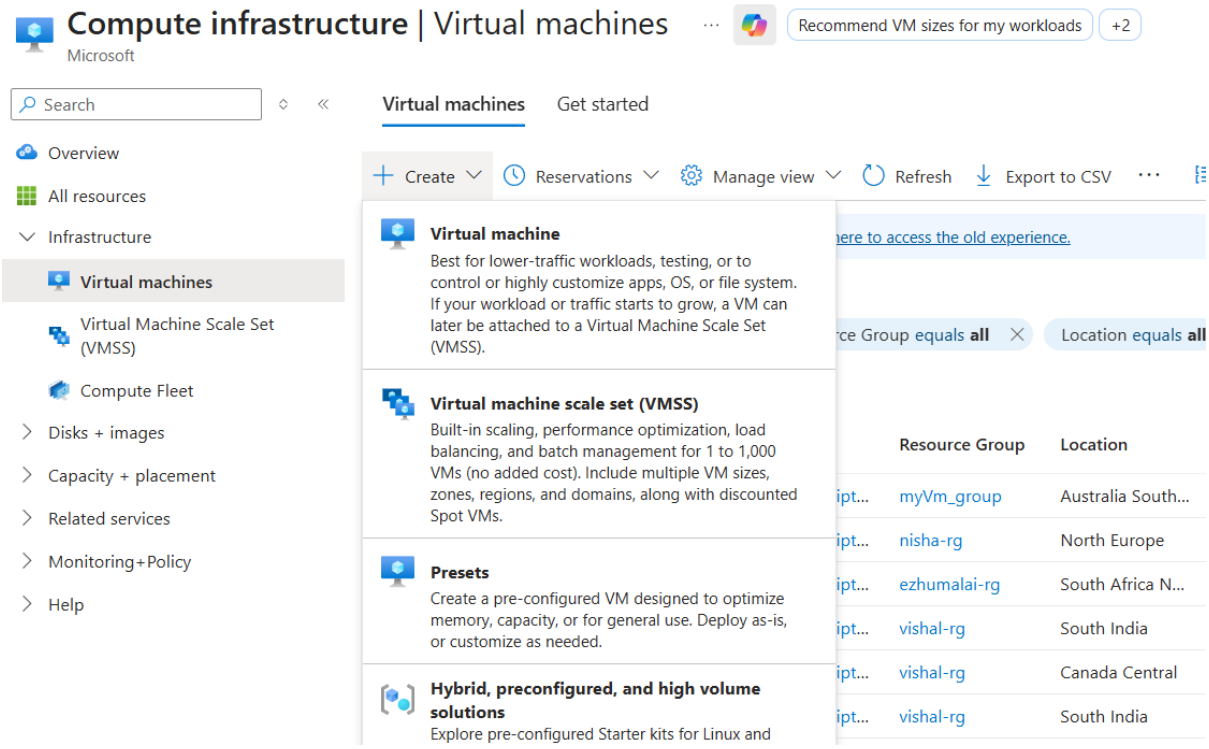
✅ **Allow inbound ports**

Select:

- ☑ **SSH (22)**

- ☑ **HTTP (80)**

Click **Next**



Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * ⓘ — Azure subscription 1

Resource group * ⓘ — dhanush-rg

Create new

Instance details

Virtual machine name * ⓘ — azureuser-vm

Region * ⓘ — (Asia Pacific) Australia East

Deploy to an Azure Extended Zone

Availability options ⓘ — No infrastructure redundancy required

Security type ⓘ — Standard

Image * ⓘ — AlmaLinux OS 9 - x64 Gen2 (free services eligible)

See all images | Configure VM generation

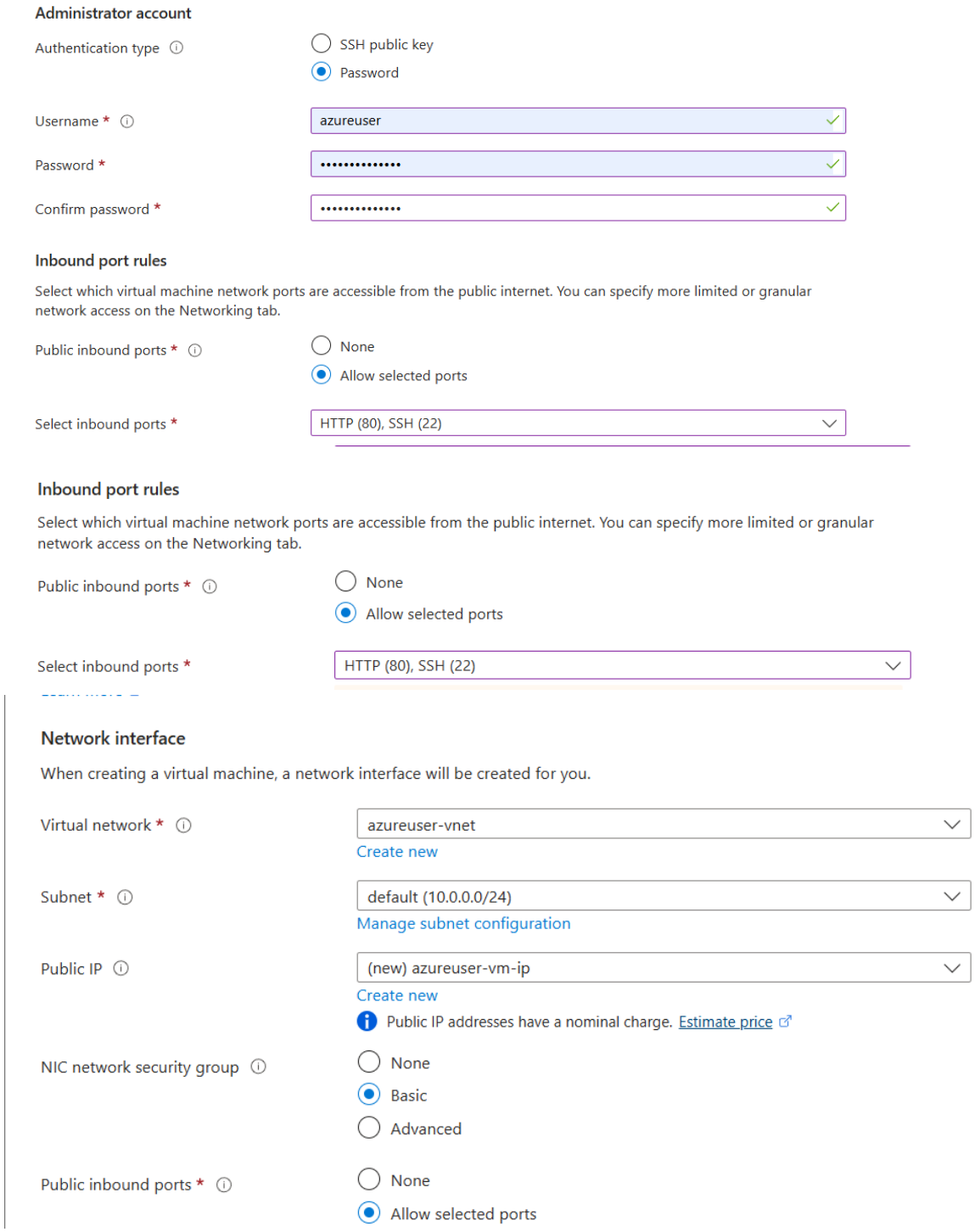
This image is compatible with additional security features. Click here to swap to the Trusted launch security type.

VM architecture ⓘ

- Arm64
- x64

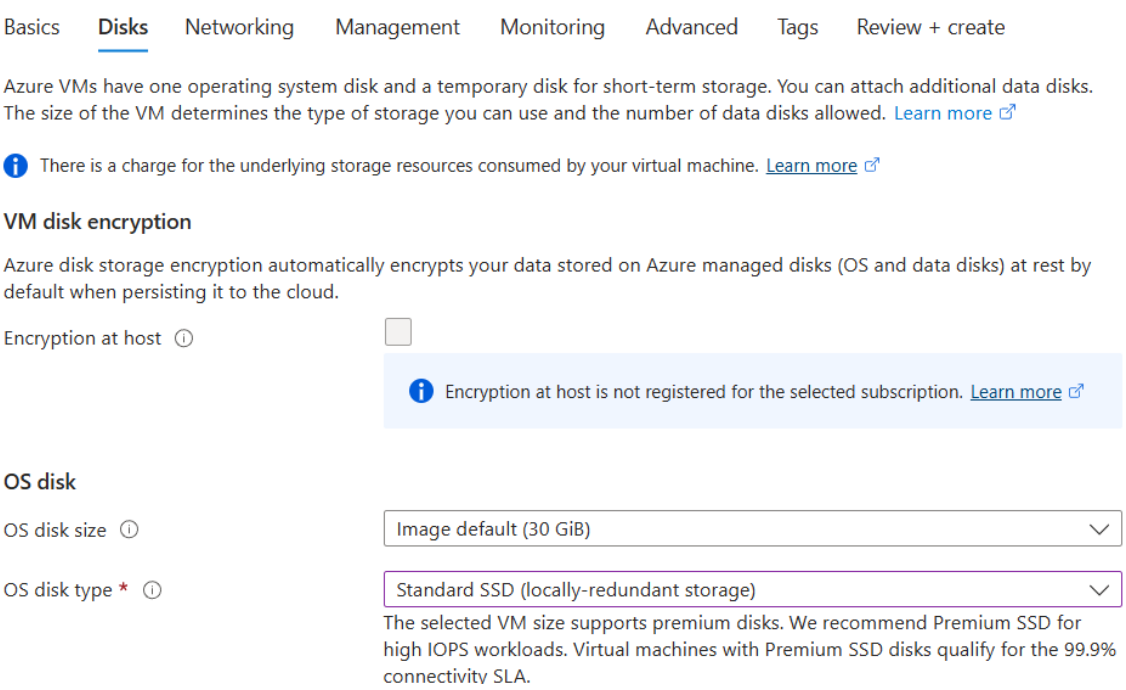
Arm64 is not supported with the selected image.

Run with Azure Spot discount ⓘ



Disks Tab

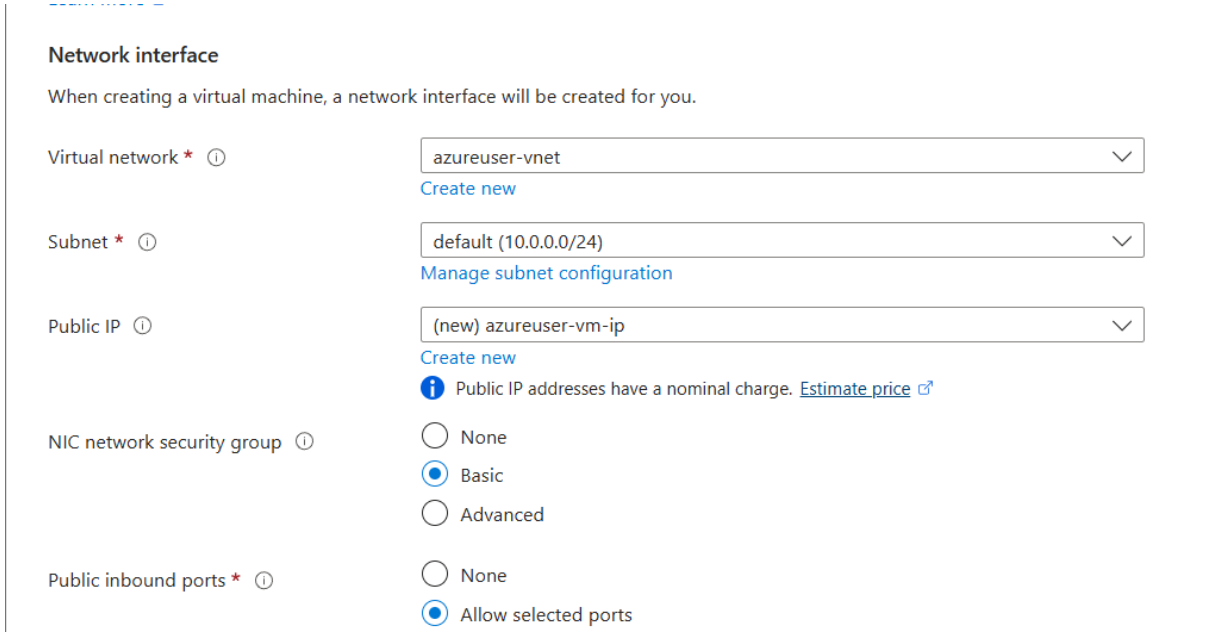
- OS disk type: Standard SSD



- Click **Next**

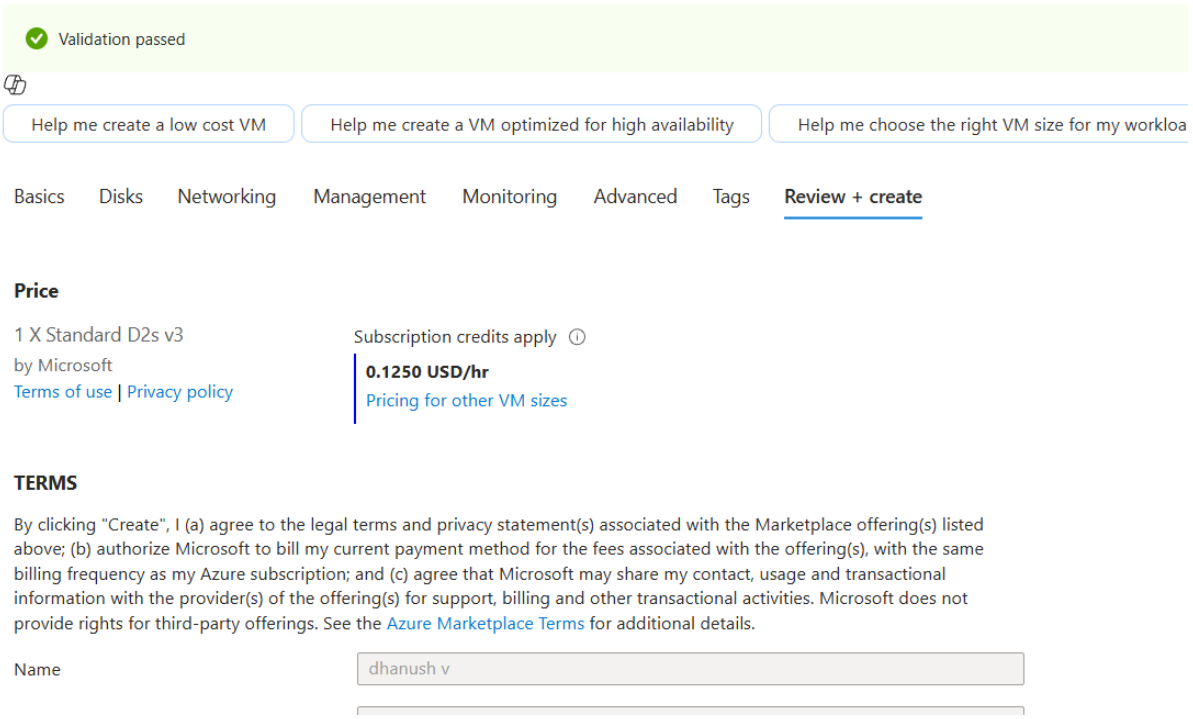
Networking Tab

- Virtual network: studentname-vnet (new)
- Subnet: studentname-subnet
- Public IP: New
- NIC network security group: Basic



👉 **No need to add port rules here** (already done in Basics)

Click **Review + Create → Create**



Step 3: Connect to VM

1. Open studentname-vm
2. Click **Connect → SSH**
3. Use Azure Cloud Shell or local terminal

Step 4: Configure VM (Inside VM)

Create directories

sudo mkdir /d1 /d2 /d3 /d4 /d5

Verify:

ls /

Install required packages

RHEL / CentOS / Rocky

sudo yum install -y httpd wget zip net-tools

Ubuntu

sudo apt update

sudo apt install -y apache2 wget zip net-tools

Start and enable HTTP service

sudo systemctl enable httpd

sudo systemctl start httpd

Create test web page

echo "Welcome from studentname custom image VM" | sudo tee /var/www/html/index.html

Verify:

curl localhost

Browser:

http://<VM-Public-IP>

```
C:\Users\DHANUSH>ssh azureuser@20.37.218.162
The authenticity of host '20.37.218.162 (20.37.218.162)' can't be established.
ED25519 key fingerprint is SHA256:U02tKWNHEGH2Z0hBbg6/G/ic3HEk0uuRHTRXQPEoNsc.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '20.37.218.162' (ED25519) to the list of known hosts.
azureuser@20.37.218.162's password:
[azureuser@azureuser-vm ~]$ sudo -i
[root@azureuser-vm ~]# mk d1 d2 d3 d4 d5
-bash: mk: command not found
[root@azureuser-vm ~]# mkdir d1 d2 d3 d4 d5
[root@azureuser-vm ~]# ls -l
total 0
drwxr-xr-x. 2 root root 6 Dec 25 16:50 d1
drwxr-xr-x. 2 root root 6 Dec 25 16:50 d2
drwxr-xr-x. 2 root root 6 Dec 25 16:50 d3
drwxr-xr-x. 2 root root 6 Dec 25 16:50 d4
drwxr-xr-x. 2 root root 6 Dec 25 16:50 d5
[root@azureuser-vm ~]# pwd
/root
```

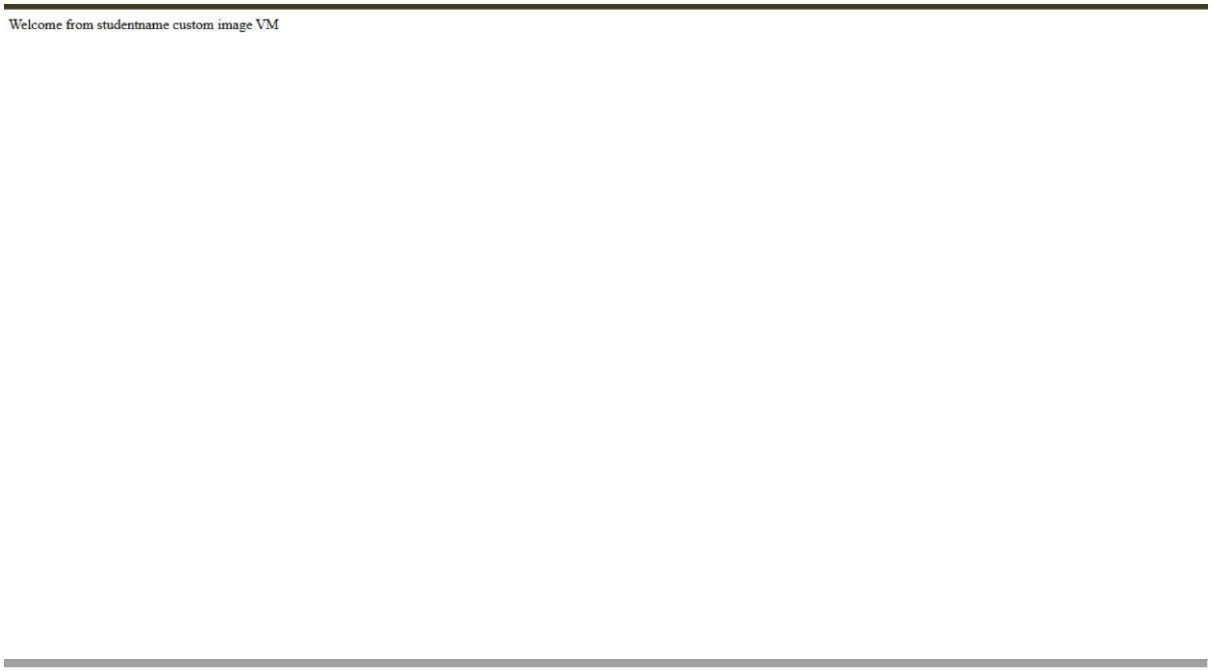
```
[root@azureuser-vm ~]# yum install httpd wget net-tools zip -y
Last metadata expiration check: 0:00:12 ago on Thu 25 Dec 2025 04:51:46 PM UTC.
Dependencies resolved.
================================================================================
 Package                  Arch        Version                      Repository      Size
================================================================================
Installing:
 httpd                    x86_64      2.4.62-7.el9_7.3             appstream       45 k
 net-tools                x86_64      2.0-0.64.20160912git.el9     baseos         294 k
 wget                     x86_64      1.21.1-8.el9_4               appstream      768 k
 zip                      x86_64      3.0-35.el9                   baseos         263 k
Installing dependencies:
 almalinux-logos-httpd    noarch      90.6-2.el9                   appstream       18 k
 apr                      x86_64      1.7.0-12.el9_3               appstream      122 k
 apr-util                 x86_64      1.6.1-23.el9                 appstream       94 k
 apr-util-bdb             x86_64      1.6.1-23.el9                 appstream       12 k
 httpd-core               x86_64      2.4.62-7.el9_7.3             appstream      1.4 M
 httpd-filesystem         noarch      2.4.62-7.el9_7.3             appstream       12 k
 httpd-tools              x86_64      2.4.62-7.el9_7.3             appstream       79 k
 mailcap                  noarch      2.1.49-5.el9                 baseos          32 k
 unzip                    x86_64      6.0-59.el9                   baseos         179 k
Installing weak dependencies:
 apr-util-openssl         x86_64      1.6.1-23.el9                 appstream       14 k
 mod_http2                x86_64      2.0.26-5.el9                 appstream      163 k
 mod_lua                  x86_64      2.4.62-7.el9_7.3             appstream       59 k

Transaction Summary
================================================================================
Install  16 Packages

Total download size: 3.5 M
Installed size: 11 M
```

```
[root@azureuser-vm ~]# cd /var/www/html/
[root@azureuser-vm html]# ls
[root@azureuser-vm html]# ll
total 0
[root@azureuser-vm html]# vi index.html
[root@azureuser-vm html]# cat index.html
echo "Welcome from studentname custom image VM" | sudo tee /var/www/html/index.html
[root@azureuser-vm html]# systemctl status httpd
○ httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabled)
     Active: inactive (dead)
       Docs: man:httpd.service(8)
[root@azureuser-vm html]# systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[root@azureuser-vm html]# systemctl status httpd
○ httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
     Active: inactive (dead)
       Docs: man:httpd.service(8)
[root@azureuser-vm html]# curl ifconfig.me
20.37.218.162[root@azureuser-vm html]#
```

Out put :



Step 5: Capture VM Image (GUI)

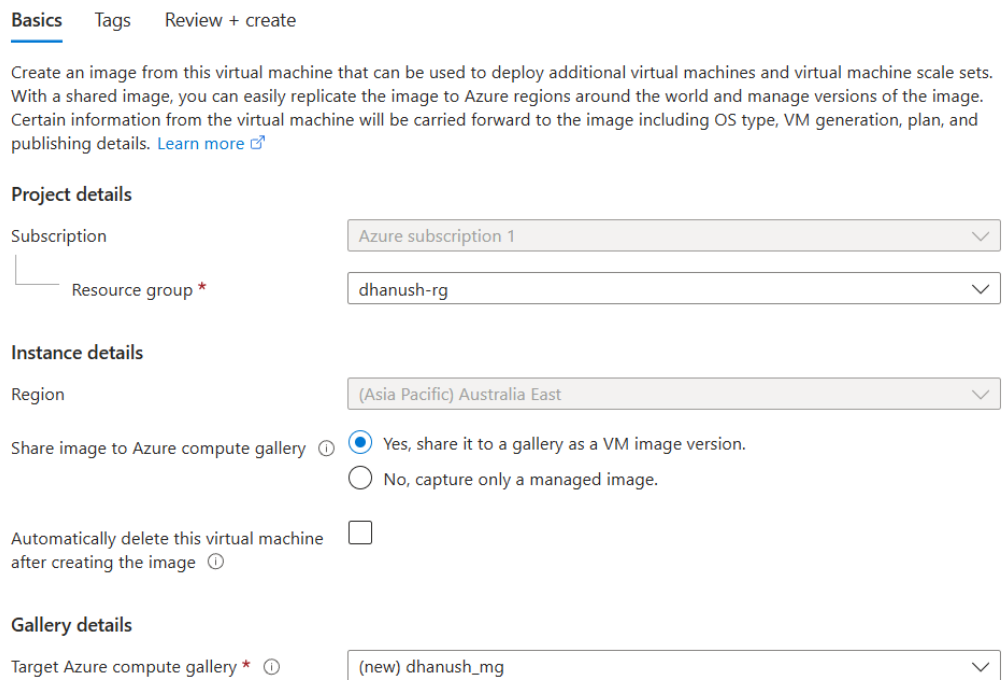
✅ *No manual deprovision required*

1. Open studentname-vm
2. Click **Capture**
3. Fill details:
    - Resource group: studentname-rg
    - Share image to Azure Compute Gallery: **Yes**
    - Automatically delete VM after creating image: ❌ No
    - Gallery: **Create new**

        - Name: studentname-acg
    - Image definition: **Create new**
        - Name: studentname-img-def
        - OS type: Linux
        - OS state: **Generalized**
    - Image version: 1.0.0
    - Target region: East US

4. Click **Review + Create → Create**

⏳ Wait until image creation completes.



Create a VM image definition

VM image definition name * : dhanush-img

OS type : Linux / Windows

VM generation : Gen 1 / Gen 2

Security type : Standard

VM architecture : x64 / Arm64

Higher storage performance with NVMe

Hibernation supported

Accelerated networking

Publisher * : almalinux

Offer * : almalinux-x86_64

SKU * : 9-gen2

Ok | Cancel | Give feedback

Gallery details

Target Azure compute gallery * : (new) dhanush_mg

Create new

Operating system state :

- Generalized: VMs created from this image require hostname, admin user, and other VM related setup to be completed on first boot
- Specialized: VMs created from this image are completely configured and do not require parameters such as hostname and admin user/password

⚠ Capturing a virtual machine image will make the virtual machine unusable. This action cannot be undone.

Target VM image definition * : (new) dhanush-img

Create new

Version details

Version number * : 1.0.0

Exclude from latest

End of life date : 12/27/2025

Block deletion before End of life

Exclude from latest ⓘ ☐

End of life date ⓘ 12/27/2025

Block deletion before End of life ⓘ ☐

Lock deleting Replicated Locations ⓘ ☑

Shallow replication ⓘ ☐

Replication

A VM image version can be replicated to different regions depending on what makes sense for your organization. One example is to always replicate the latest image in multiple regions while all older versions are only available in 1 region. This can help save on storage costs for VM image versions.

Default storage sku ⓘ Zone-redundant

Default replica count * ⓘ 1

Target regions	Replica count	Storage sku	Exclude from latest
(Asia Pacific) Austra...	1	Zone-redundant	☐
(US) East US	1	Standard HDD LRS	☐

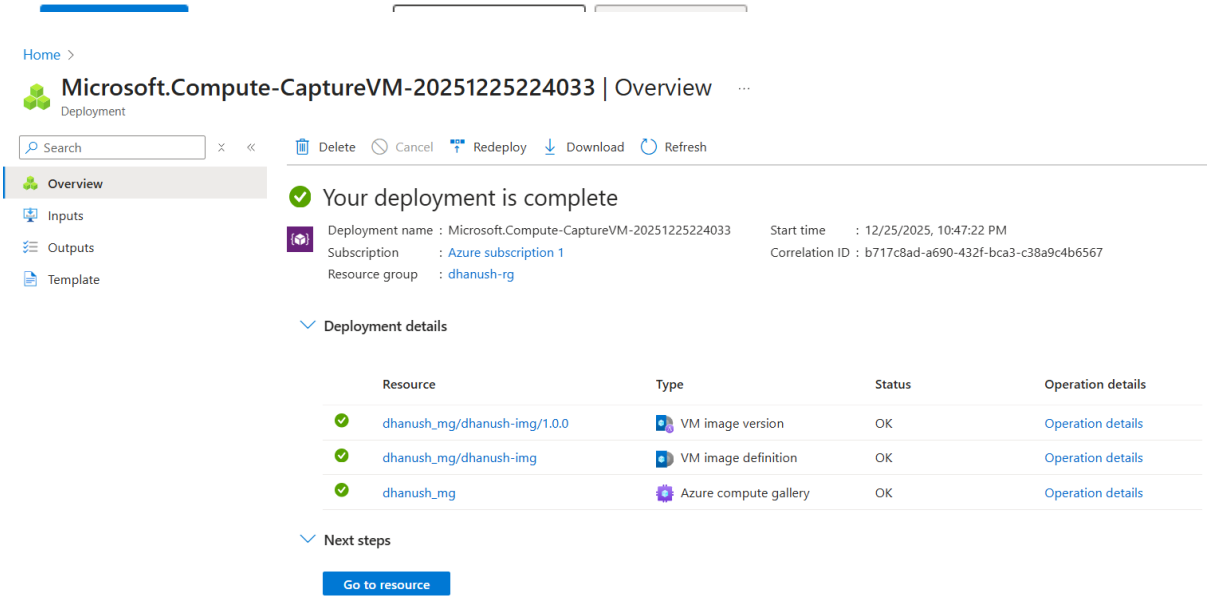
Create an image

Validation passed

Basics Tags **Review + create**

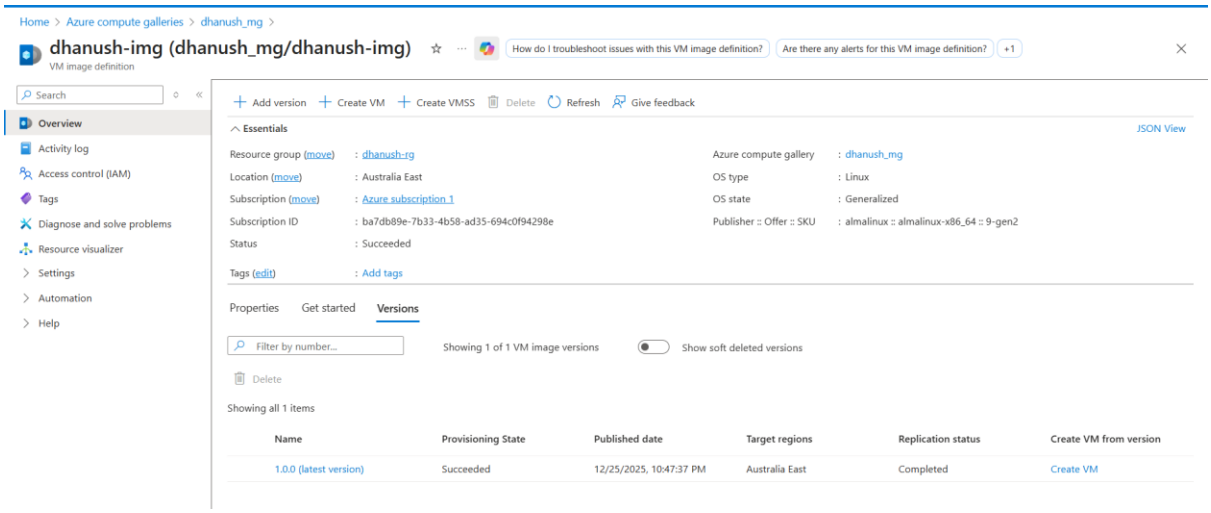
Basics

Subscription	Azure subscription 1
Resource group	dhanush-rg
Region	Australia East
Share image to Azure compute gallery	Yes
Automatically delete this virtual machine after creating the image	No
Azure compute gallery	(new) dhanush_mg
Operating system state	Generalized
Target VM image definition	(new) dhanush-img
Version number	1.0.0
Source virtual machine	azureuser-vm
Exclude from latest	No
End of life date	2025-12-26
Lock deleting Replicated Locations	Yes
Shallow replication	No



Step 6: Verify Azure Compute Gallery

1. Search → **Azure Compute Galleries**
2. Open studentname-acg
3. Confirm:
    - Image Definition exists
    - Image Version 1.0.0 status = **Succeeded**



Step 7: Create VM from Image (GUI)

1. Open **Image version 1.0.0**
2. Click **Create VM**

Basics

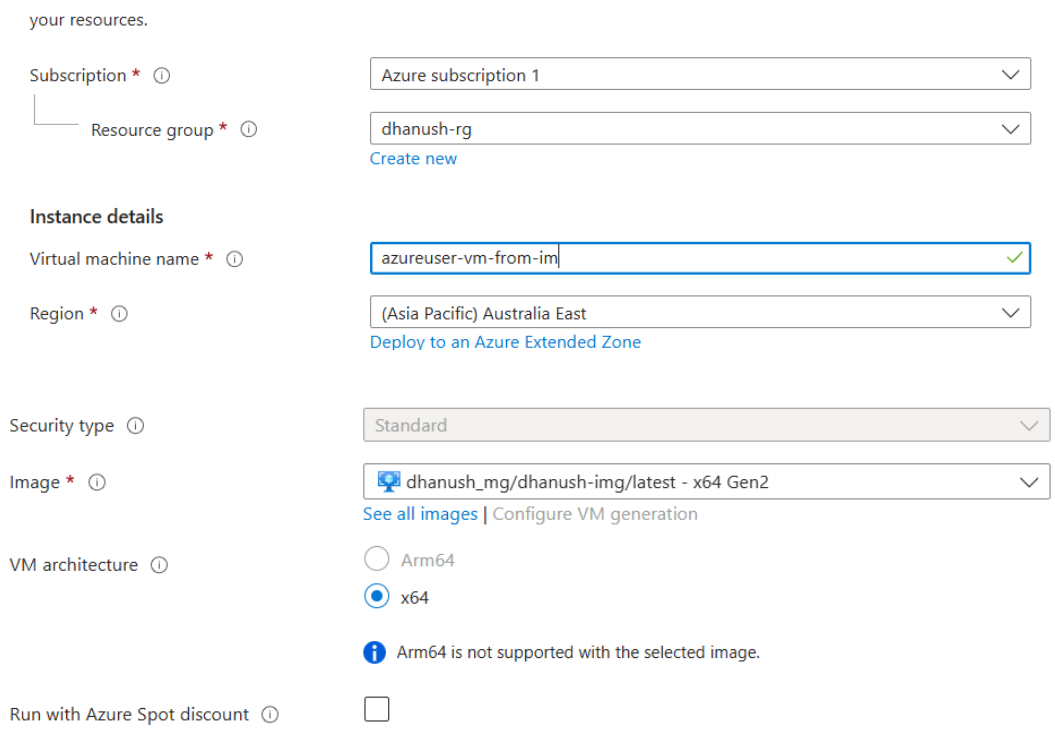
- Resource group: studentname-rg
- VM name: studentname-vm-from-img
- Size: Standard_B2s
- Username: azureuser
- SSH key: Use existing key

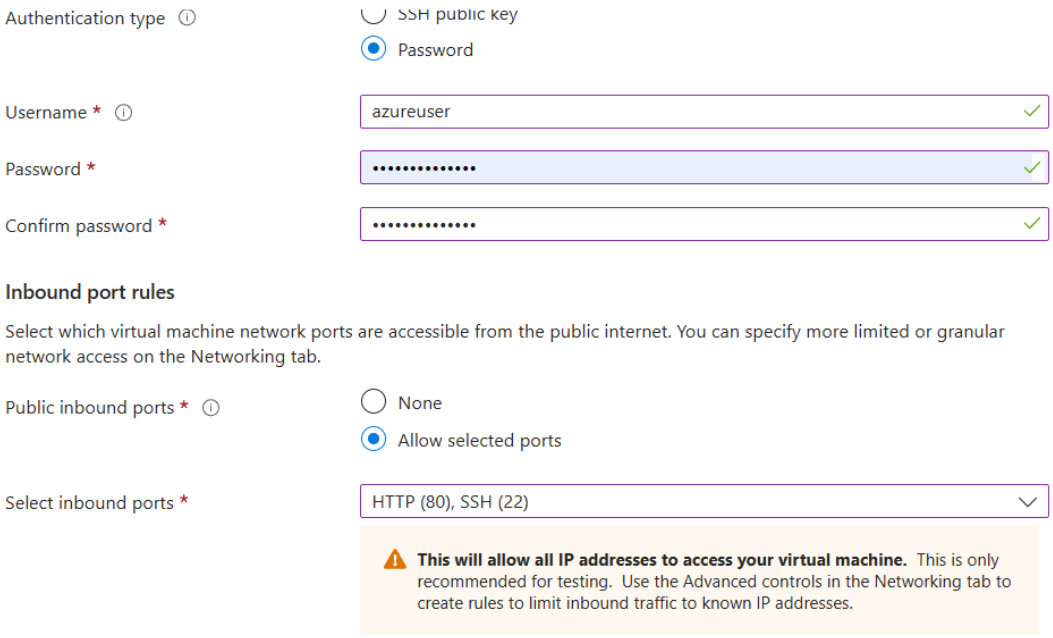
✅ Allow inbound ports

Select:

- ☑ **SSH (22)**
- ☑ **HTTP (80)**

Click **Review + Create → Create**





Step 8: Validate New VM

SSH into new VM

ssh azureuser@<NEW-VM-PUBLIC-IP>

Validate directories

ls /

Expected:

d1 d2 d3 d4 d5

Validate installed packages

httpd -v

wget --version

zip -v

ifconfig

Validate HTTP service

systemctl status httpd

curl localhost

Browser:

http://<NEW-VM-PUBLIC-IP>

output :

Welcome from studentname custom image VM